

Staffing Shortages and Emergency Department Closures in France

Mid-Term Update

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Overview

1. Research Questions
2. Data Sources
3. Descriptive Statistics
4. EDCD
5. Next Steps

Research Questions

Main question

- How do staffing levels in emergency departments relate to the occurrence of partial or full closures, or regulated access, in 2024 in France?

Secondary questions

- How are characteristics such as legal status, geographic location, or hospital size associated with partial or full closures?
- Are hospitals with low doctor-to-visit ratios more likely to experience closures or regulated access? Are départements with low emergency departments-to-visit ratios more likely to experience closures or regulated access?
- How has staffing distribution evolved (2014 vs 2024)?

Data Sources

Administrative datasets

- SAE — staffing and activity indicators
- Enquête Urgences 2023 — detailed staffing snapshot

New dataset under construction

- EDCD: Emergency Department Closures Database

Enquête Urgences: Dataset

Activity

- Snapshot of staffing (morning/evening/night)

Key variables: Ratios

- Patient to staff – includes non medical staff (e.g., receptionists, stretcher-bearers)
- Patient to medical staff
- Non-permanent to total staff
- Non-permanent to medical staff

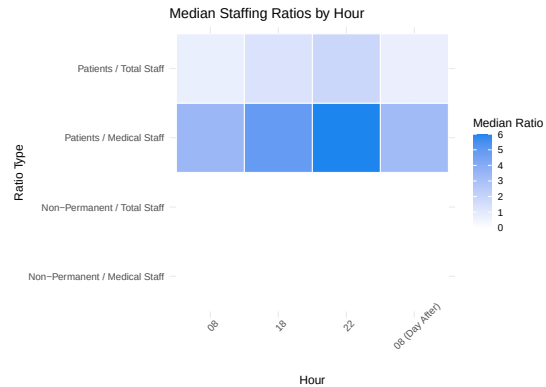
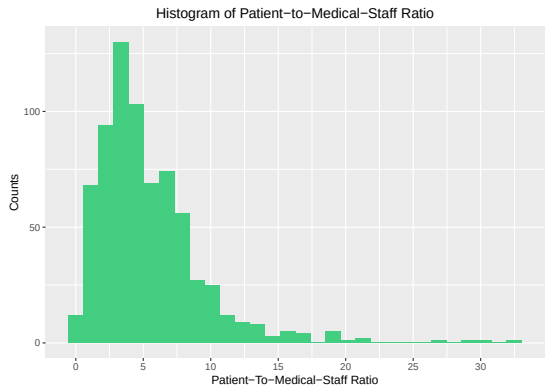
Usefulness

- Validating staffing measures.
- Understanding daily patterns.

Limitations

- Limited external validity – June 13, 2023

Enquête Urgences: Preliminary Descriptive Statistics



SAE: Dataset

Activity

- Annual number of emergency visits – 2014 and 2024

Staffing

- Physicians – salaried and liberal
- Junior doctors and interns

Data cleaning: Steps and challenges

- Merge multiple tables – identifiers, activity, personnel, etc.
- Deal with outliers – possible survey filling errors
- Compare 2014 and 2024
 - SAE 2014 lacks several emergency-specific variables
 - Find comparable variables – e.g., Hours of *salaried* emergency physicians (2024) = Hours of *all* (salaried and non-) emergency physicians

Descriptive Statistics: Work in Progress

Current focus

- Distribution of doctor-to-visit ratios
- Distribution of ED-to-visit ratios
- Regional heterogeneity
- Hospital characteristics

Comparison

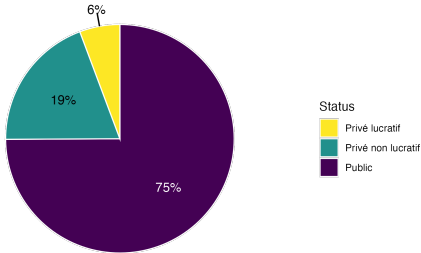
- 2014 vs 2024 (cross-sectional)

Note

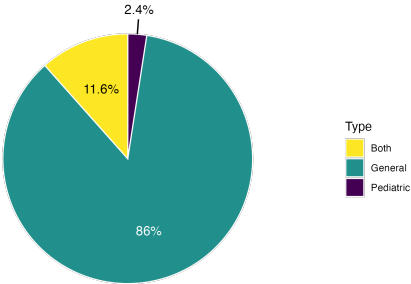
- Quality checks ongoing.
- Results preliminary.

General Description of the Dataset

Juridical Status of the Emergency Services

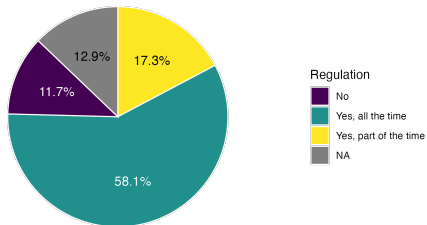


Type of the Emergency Service

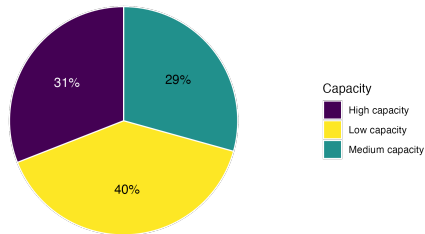


General Description of the Dataset

Regulation of the Emergency Services

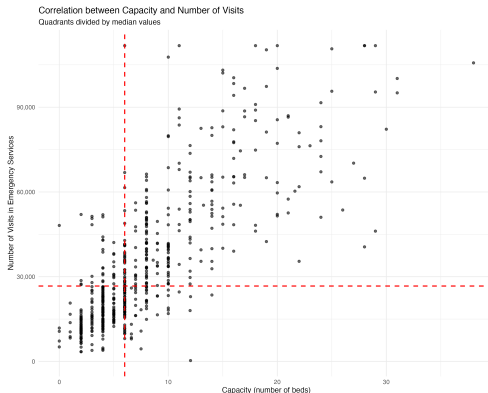


Capacity of the Emergency Services



General Description of the Dataset

- Plot the correlation between the number of visits and the capacity (number of beds), to isolate the emergency services that exhibit a high number of patients for a low capacity.



- Focusing on the top-left quadrant.

General Description of the Dataset

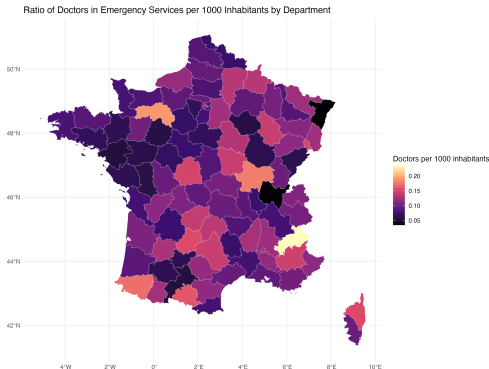
- Explore the distribution of the top-left quadrant hospitals.

Distribution of the Overloaded Hospitals						
variable	category_label	pct_overloaded	pct_global	pct_diff	p_value	representation
juridical_status	Privé lucratif	15.15	5.70	9.45	<0.001	Over-represented
juridical_status	Privé non lucratif	36.36	19.38	16.98	<0.001	Over-represented
juridical_status	Public	48.48	74.92	-26.43	<0.001	Under-represented
regulation	No	12.12	11.73	0.39	0.0826	Over-represented
regulation	Part-time	15.15	17.26	-2.11	0.0826	Under-represented
regulation	Full-time	45.45	58.14	-12.69	0.0826	Under-represented
regulation	NA	27.27	12.87	14.41	0.0826	Over-represented
emergency_type	General	96.97	85.99	10.98	0.1706	Over-represented
emergency_type	Both	3.03	11.56	-8.53	0.1706	Under-represented
emergency_type	Pediatric	0.00	2.44	-2.44	0.1706	Under-represented

- Only the difference in distribution across the juridical sector is significant: private and semi-private hospitals are overly represented across overloaded hospitals.

Doctors to population and patients ratios

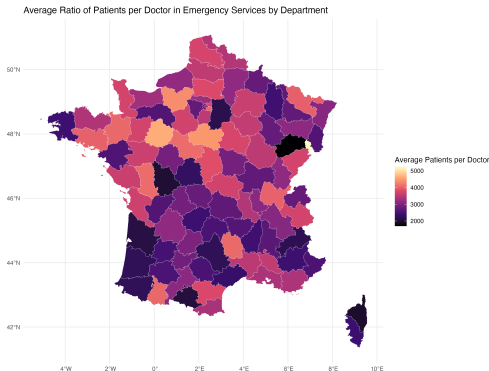
- Explore the ratio of emergency doctors per 1000 inhabitants.



- Bas-Rhin, Ain, and Tarn-et-Garonne have the lowest ratio of doctors per 1000 inhabitants (respectively 3.45, 3.50, and 4.56 per 100 000 inhabitants).

Doctors to population and patients ratios

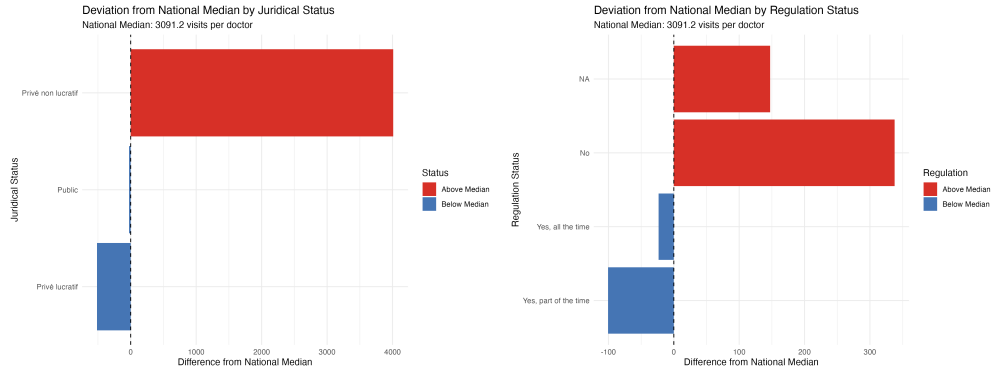
- Explore ratio of patients to doctor.



- Sarthe, Loiret, and Eure have the highest ratio of patients per doctor (around 4500 per year, ie around 21 per day).

Doctors to population and patients ratios

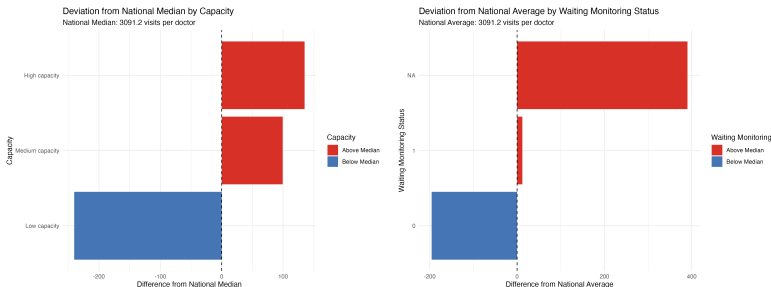
- Explore the deviation from the median for this ratio across juridical status and regulation system.



- Semi-private and non-regulated emergency rooms exhibit a patients-to-doctor ratio above the median.

Doctors to population and patients ratios

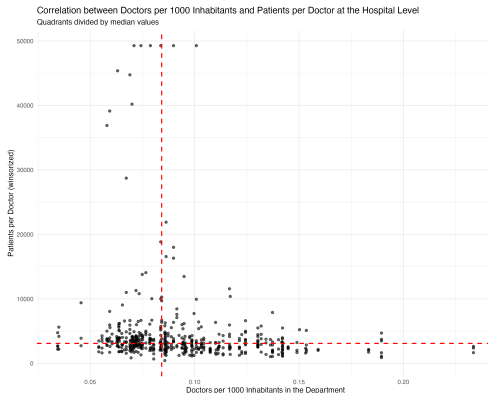
- Explore the deviation from the median for this ratio across capacity and waiting monitoring system.



- High and medium capacity emergency rooms exhibit a patients-to-doctor ratio above the median. The results on the waiting monitoring system are not conclusive.

Doctors to population and patients ratios

- Divide the previous scatterplot in four quadrants based on the medians, and study the top-left corner, ie the hospitals with a low doctors to inhabitant ratio and a high patients to doctor ratio.

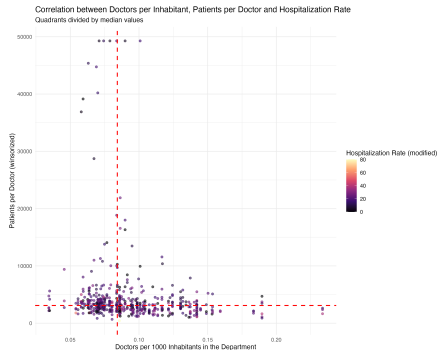


Doctors to population and patients ratios

- Further analysis: Explore the distribution across the type, regulation, and juridical sector dimensions (Chi-square table).
- Further analysis: Explore the deviation from the median across the same variables.

Severity analysis

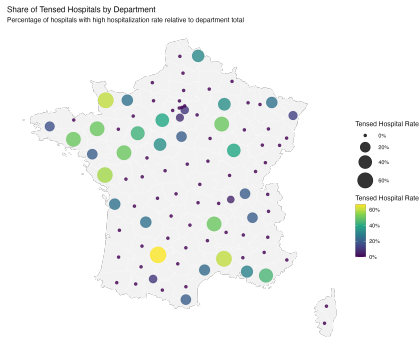
- Explore the correlation between the doctor-to-population ratio, the patients-to-doctor ratio, and the hospitalization rate.



- Isolate the hospitals in the top left corner (low doctors per inhabitant, high patients per doctor) and check if they are also the ones with the highest hospitalization rates.

Severity analysis

- Explore the distribution of tensed hospitals.



- Further analysis: Explore the distribution across the type, regulation, and juridical sector dimensions (Chi-square table), the deviation from the median across the same variables.

Further Analysis Ideas: Personnel

- Analyze the ratio of interns and junior doctors over the total medical staff.
First, in emergency services that have a high ratio, the seniors also have a supervisory burden. Second, since interns and junior doctors move, some spots may be vacant, meaning that services that highly rely on interns show a structural fragility.
- Joint Personnel: Analyze the joint personnel between the emergency structure and the SMUR to identify hospitals where the same staff must cover both the ER and mobile units (SMUR).

Emergency Department Closures Database (EDCD)

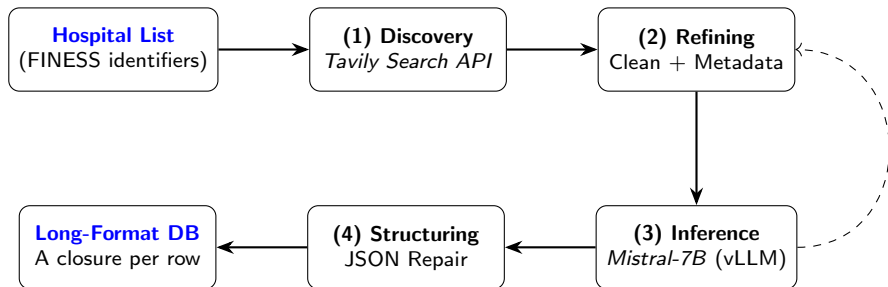
Objective

- Construct a reproducible, auditable database of ED closures and regulated access episodes in France, since 2023.

Sources

- **Administrative:** Arrêtés préfectoraux, ARS decisions, official government publications.
- **Media & Public Reporting (Non-administrative):** Hospital websites, local press, and social media communications.

EDCD: Pipeline



- **Auditability:** 1 Extraction = 1 Row = 1 Verifiable URL.
- We treat each URL as a unique piece of evidence.

EDCD: (1) Discovery



Function: Identify and retrieve all potentially relevant evidence of ED disruptions.

Implementation via Tavily API

- Web search across news, institutional pages, and social platforms
- HTML-to-text extraction
- Relevance scoring (*score*) to rank candidate sources
- **High-recall retrieval:** capture any mention of *fermeture* or *régulation* (recall over precision)
- 4,000 credits per month for students (approximately 2,000–4,000 searches)

EDCD: (1) Discovery

```
query = f'("{name}" OR "{keywords_nom_etab}") service des urgences fermeture régulation accès régulé'

if len(query) > 400:
    query = f'"{keywords_nom_etab}" service des urgences fermeture régulation accès régulé'

try:
    search = tavily_client.search(
        query=query,
        search_depth="advanced",
        topic="general",
        chunks_per_source=3,
        max_results=10,          # ← keep high to catch multiple events
        country="france",
        start_date="2022-06-01",
        end_date="2025-12-31"
    )

    results = search.get('results', [])
```

- Search queries designed to mimic Google-style searches
- Iteration over the 609 unique emergency departments (Salles d'Urgences) identified in the Enquête Urgences survey.
- Reduce structural bias in retrieval relevance.

EDCD: (1) Discovery

```
search_official = tavily_client.search(  
    query=query,  
    search_depth="advanced",  
    topic="general", # includes ALL source types  
    chunks_per_source=5,  
    max_results=5,  
    include_domains=["ars.sante.fr", "sante.gouv.fr"],  
    country="france",  
    start_date="2022-06-01",  
    end_date="2025-12-31"
```

```
search_unofficial = tavily_client.search(  
    query=query,  
    search_depth="advanced",  
    topic="general",  
    chunks_per_source=5,  
    max_results=5,  
    exclude_domains=["ars.sante.fr", "sante.gouv.fr"],  
    country="france",  
    start_date="2022-06-01",  
    end_date="2025-12-31"
```

- **Official sources:** High institutional credibility and legal authority, but often delayed or incomplete for short-term disruptions.
- **Unofficial sources:** Broader and more reactive coverage; hospitals and local press frequently report temporary closures before formal publication.
- **Trade-off:** Precision vs. Recall. Official sources increase precision; unofficial sources increase recall. Combining both maximizes coverage but requires careful de-duplication and cleaning.

EDCD: (1) Discovery

▶ View Retrieved Data

EDCD: (2) Refining

Function: Process raw search noise into data easy to extract.

Data Preparation

- Control information granularity to balance recall and contextual coherence.
- **Recall focus:** Sending events line-by-line to maximize coverage. Ensuring no short texts indicating closures are overlooked by the model.
- **Precision focus:** Grouping all events by hospital to the machine interpret the context—allowing the LLM to cross-reference dates and ignore conflicting "noise" to produce a single ground truth.
- Truncating long decrees while preserving legislative headers and "meat" text.
- We "staple" the `source_url` to every text snippet, ensuring every entry in the final database is 100% auditable.

EDCD: (3) Inference

Function: Extract event attributes.



Local infrastructure

- *Mistral-7B-Instruct-v0.3* served via vLLM for batch inference.

Grounded extraction

- The model must cite the supporting `source_url` for each extracted event.
- Extract only what is explicitly stated in the text (temperature = 0).

Output constraints:

- Dates in YYYY-MM-DD.
- `closure_type` = {Full Closure, Temporary Closure, or Regulated Access}

EDCD: (2) Refining and (3) Inference: test 1

Arrêté n°2024-17-0862

Portant autorisation de réguler temporairement l'accès aux urgences du centre hospitalier de Bourgoin-Jallieu

La directrice générale de l'agence régionale de santé Auvergne-Rhône-Alpes

Vu le code de la santé publique, notamment ses articles L. 1432-2, L. 6122-1, L. 6122-8, R. 6122-25, R. 6122-41, R. 6123-1 à R. 6123-32-11;

Vu le décret du 19 avril 2023 portant nomination de madame Cécile COURREGES en qualité de directrice générale de l'agence régionale de santé Auvergne-Rhône-Alpes;

Vu l'arrêté n°2016-0714 du 21 mars 2016 portant renouvellement tacite d'autorisations d'activités de soins de médecine d'urgence du centre hospitalier de Bourgoin-Jallieu;

Vu l'arrêté du 2 juillet 2024 relatif à la régulation temporaire de l'accès aux urgences;

```
{
  "query": "Arrêté \"Centre Hospitalier de Bourgoin-Jallieu\" urgences réguler temporairement",
  "results": [
    {
      "url": "https://www.prefectures-regions.gouv.fr/intercontenu/telechargement/123675/916331/fle/31decembre2024_n-2_recueil-84-2024-378-recueil-des-actes-administratifs-special.pdf",
      "title": "RECUEIL DES ACTES ADMINISTRATIFS SPÉCIAL N° 84-...",
      "content": "de la région Auvergne - Rhône-Alpes. Fait à Lyon, le 30 décembre 2024 La directrice générale de l'agence régionale de santé Auvergne-Rhône-Alpes Cécile COURREGES Arrêté n°2024-17-0862 Portant autorisation de réguler temporairement l'accès aux urgences du centre hospitalier de Bourgoin-Jallieu La directrice générale de l'agence régionale de santé Auvergne-Rhône-Alpes Vu le code de la santé publique, notamment ses articles L. 1432-2, L. 6122-1, L. 6122-8, R. 6122-25, R. 6122-41, R. 6123-1 à R. 6123-32-11; Vu le décret du 19 avril 2023 portant nomination de madame Cécile COURREGES en qualité de directrice générale de l'agence régionale de santé Auvergne-Rhône-Alpes; Vu l'arrêté n°2016-0714 du 21 mars 2016 portant renouvellement tacite d'autorisations d'activités de soins de médecine [...] tacite d'autorisations d'activités de soins de médecine d'urgence du centre hospitalier de Bourgoin-Jallieu; Vu l'arrêté du 2 juillet 2024 relatif à la régulation temporaire de l'accès aux urgences; Vu la décision 2024-23-0064 en date du 03 décembre 2024 portant délégation de signature de la Directrice générale de l'ARS Auvergne-Rhône-Alpes; Vu l'avis consultatif n°2024-15 du 23 décembre 2024 de la section Urgences chargée d'émettre un avis pour les activités de médecine d'urgence du comité consultatif d'allocation des ressources prévu à l'article R. 162-29 du code de la sécurité sociale; Vu la demande de l'établissement de bénéficier d'un renouvellement de l'autorisation de réguler temporairement la nuit l'accès aux urgences de son territoire; Considérant que tout établissement de [...] mis en œuvre par le centre hospitalier, l'établissement ne parvient pas à réunir les effectifs nécessaires à l'accueil des urgences sans régulation préalable; Considérant que, dans ce contexte il y a lieu de prioriser l'accueil des patients au sein de la structure des urgences, de préserver les capacités optimales de prise en charge des urgences vitales et graves des structures mobiles d'urgence et de réanimation, d'assurer une sécurité des soins et d'éviter la saturation des urgences; ARRÊTE Article 1er : Le présent arrêté prend effet pour 3 mois à compter de sa date de signature; le centre hospitalier de Bourgoin-Jallieu est autorisé à réguler l'accès à sa structure des urgences entre 20h00 et 08h30. Article 2 : L'accès à la structure des urgences s'opérera par une régulation. 'score': 0.92442214, 'raw_content': None, 'url':
    }
  ]
}
```

```
prompt = """
You are an information extraction system.

Task: Extract the requested fields from the text below.
Rules:
- Return ONLY valid JSON.
- If a field is missing, use null.
- Do not add fields.
- Do not explain.

Fields: {fields}

Text:
\"\"\"{resp}\"\"\"
"""

outputs = llm.generate([prompt], sampling_params=sampling_params)
print(outputs[0].outputs[0].text)
```

```
{
  "hospital_name": "Centre Hospitalier de Bourgoin-Jallieu",
  "authorization_type": "regulation",
  "start_date": "2024-08-02",
  "duration": "3 months",
  "hours_regulated": "20:00 to 08:30",
  "authority": "Agence Régionale de Santé Auvergne-Rhône-Alpes",
  "source_location": "Lyon"
}
```

EDCD: (3) Inference: test 2

```
def build_final_prompt(row):
    today = datetime.now().strftime("%Y-%m-%d")
    return f"""<s>[INST] You are a specialized data extractor for the French Ministry of Health (DREES).
    Today's date is: {today}. Use it to resolve relative dates.

    TASK: Extract all Emergency Room disruptions for: {row['hospital_name']}.
    Include: Full Closure, Temporary Closure, or Regulated Access

    FORMAT: Return ONLY a valid JSON list of objects.
    Keys:
    - "start_date" (YYYY-MM-DD)
    - "end_date" (YYYY-MM-DD or "Ongoing")
    - "closure_type" (Full Closure, Temporary Closure, or Regulated Access)
    - "source_url" (The EXACT URL provided in the [SOURCE_X] tag where you found the info)

    CONTEXT:
    {row['full_context']} [/INST]"""
```

⇒ **Hallucinations**

EDCD: (4) Structuring

Function: enforce a consistent structure and flag unverifiable outputs.

JSON repair (non-semantic)

- Fix minor formatting issues (e.g., missing brackets), without altering content.

Schema enforcement

- Required fields: `start_date`, `end_date`, `closure_type`, `source_url`.
- Validity checks: date format, date ordering, categories in allowed sets.
- Traceability check: `source_url` must match the event it is referring to.

Handling missing information

- If key information is not stated (e.g., year), label as `Unknown` (no imputation).
- Ambiguous/low-confidence cases are flagged for manual review.

Result: structured, auditable event table ready for analysis in R/Stata.

Next Steps

SAE:

1. Analyze interns personnel.
2. Compare 2024 and 2014.

EDCD:

1. Refine the extraction prompt for *Mistral* (inference).
2. Finalize data cleaning and post-processing (structuring).
3. Conduct consistency and quality checks using RPU data.

Regression analysis

Thank you

Questions?